FEChAR Tutorial

The heating-ramp problem

Folder `Tutorials/ConvectiveRamp_Bprime` | Earth, Mars, Venus, Titan

What this tutorial does

FEChAR simulates how a heat-shield sample (here TACOT, a carbon-phenolic material) heats up and burns away. In the **ramp problem** the front face is heated by a short pulse: the heating rises to a plateau, stays there, then drops, and the sample cools.

The same problem can be run in four atmospheres, **Earth, Mars, Venus or Titan**, by changing one word.

The tutorial has three parts:

1. **Run** the simulation (Part 1).
2. **Post-process** the results with one script (Part 2).
3. **Change** the ramp or the heating level (Part 3).

The heating pulse

The boundary-layer heating conditions are specified by two quantities: the mass-transfer coefficient, $\rho_e u_e C_H$ [kg m^{-2} s^{-1}], and the recovery enthalpy, H_r [J kg^{-1}]. Both quantities follow the same time-dependent pulse, as shown in Figure 1. The surface mass loss due to char oxidation is computed using the B' tables corresponding to the selected atmosphere. The ramp-heating problem is based on the benchmark presented at the Ablation Workshop by Lachaud et al. [3]. The predicted in-depth temperature and mass-flux histories are compared with the PATO results [2] in Figure 2.

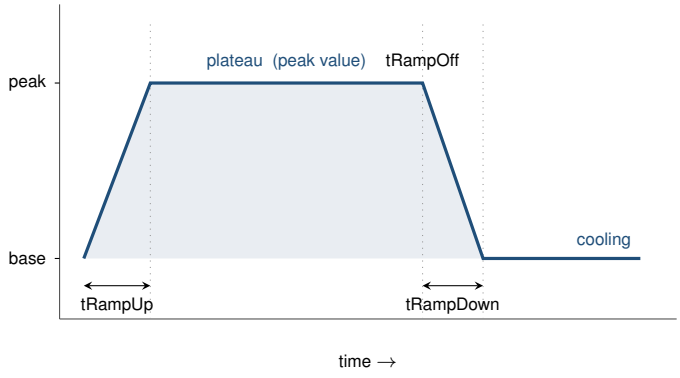


Figure 1. The heating pulse. Each quantity rises from its base value (zero for $\rho_e u_e C_H$ and H_r) to its peak, stays there, then returns to the base value.

Table 1. Pulse settings (in `params.idp`).

Setting	Flag	Default	Meaning
`tRampUp`	`-tup`	0.1 s	time to rise to the plateau
`tRampOff`	`-toff`	60 s	time when the ramp-down starts (end of plateau)
`tRampDown`	`-tdown`	0.1 s	duration of the ramp-down
`reuechPeak`	–	0.3 kg/m^2/s	plateau of $\rho_e u_e C_H$
`heatbc`	`-heat`	high	plateau of H_r: low 1.5, intermediate 13.3, high 25 MJ/kg

The equations are in `docs/THEORY.md`; they are not needed to follow this tutorial.

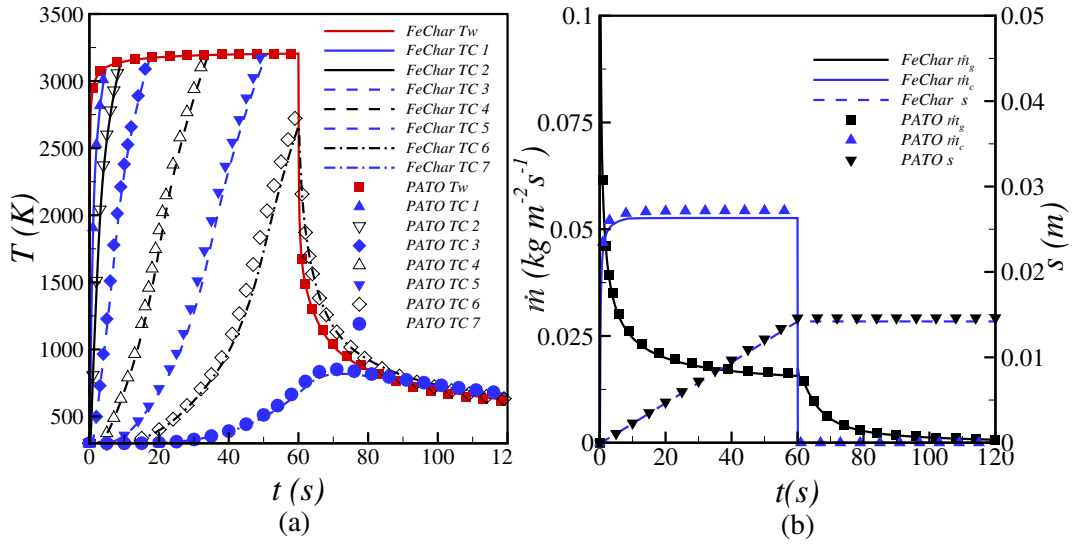


Figure 2. Validation against Benchmark Case 2 of the Ablation Workshop [3].

Before you start

Software. FreeFEM++[1] 4.x *with the `msh3` plugin* (the official installers from freefem.org include it) on Linux, macOS or WSL. For post-processing: Python 3 with NumPy and Matplotlib.

> **Ubuntu / Debian.** The `freefem++` package alone has no plugins and `load "msh3"` fails. Install the plugin package too and point FreeFEM to it:
>
> ```
> sudo apt install freefem++ libfreefem++
> export FF_LOADPATH=/usr/lib/freefem++
> ```

Where to work. Always start from inside the tutorial folder. All paths are relative.

```
FEChar1D/
  Datafolder/                  material model, B' tables, heating macros (shared)
  scripts/                     post-processing script (Python)
  Tutorials/ConvectiveRamp_Bprime/
    params.idp                 all settings: edit this file (or use command-line flags)
    main.edp                   the solver; normally not edited
    run_all_environments.sh    runs the four environments one after the other
```

Part 1: Run the simulation

Step 1. Open the folder.

```
git clone https://github.com/goutham2023/FEChAR.git
cd FEChAR/FEChar1D/Tutorials/ConvectiveRamp_Bprime
```

Step 2. Choose the environment. Set `Environment` in `params.idp`, or give `-env` on the command line. It selects the surface-chemistry table and the wall pressure.

Table 2. The wall pressure used for the four environments.

`-env`	Wall pressure
Earth	101 325 Pa
Mars	610 Pa
Venus	9.2×10^6 Pa
Titan	146 900 Pa

Step 3. Choose the heating level. `heatbc` (flag `-heat`) is `low` (1.5 MJ/kg), `intermediate` (13.3 MJ/kg) or `high` (25 MJ/kg).

Step 4. Run. Use the quick test first. The flag `-nw` switches off the graphics window. The current time is printed while the run advances.

```
# quick test: Mars, high heating (8 s, ramp-down starts at 4 s)
FreeFem++ -nw main.edp -env Mars -heat high -tend 8 -toff 4 -ele 100 -dt 0.005

# full run: uses the settings in params.idp
FreeFem++ -nw main.edp
```

Table 3. Quick test and full run.

Setting	Flag	Quick test	Full run (default)
end time	`-tend`	8 s	120 s
ramp-down starts	`-toff`	4 s	60 s
elements	`-ele`	100	500
time step	`-dt`	0.005 s	0.005 s

> **Keep dt $\leq$ 0.005 s.** A larger time step (0.01 s) made the run diverge in tests. For a faster test, reduce the end time and the number of elements, not the time step.

The results are written to a folder named `case_<Environment>_Bprime_<heat>`, for example `case_Mars_Bprime_high`. **Every run deletes and recreates this folder**, so copy results elsewhere if you want to keep them.

All four environments at once. The script `run_all_environments.sh` runs Earth, Mars, Venus and Titan in turn. Extra arguments are passed to FreeFEM; `HEAT=low` changes the heating level.

```
./run_all_environments.sh -tend 8 -toff 4 -ele 100 -dt 0.005
HEAT=low ./run_all_environments.sh -tend 8 -toff 4 -ele 100
```

Part 2: Post-process the results

Once the run has finished, one script does all the post-processing. From the tutorial folder:

```
python ../../scripts/postprocess.py case_Mars_Bprime_high
```

The script

1. extracts temperature, pressure and density histories at fixed depths and saves them in the case folder as `probes_temperature.d` `probes_pressure.dat` and `probes_density.dat`;
2. determines the positions of the virgin and char fronts and saves them as `interfaceloc.dat` in the case folder;
3. writes the plots (wall temperature, surface recession, blowing, in-depth profiles, front positions and probe histories) to the folder `figures/`.

Several case folders can be given at once; the curves are then overlaid, which is convenient for comparing environments:

```
python ../../scripts/postprocess.py case_Earth_Bprime_high case_Mars_Bprime_high
```

Useful options: `-outdir` (figure folder), `-xmax` (upper limit of the time axis), `-times` (times of the profile plot) and `-depths` (probe depths in metres; default 0.001, 0.002, 0.004, 0.008, 0.012, 0.016, 0.024). Run `python ../../scripts/postprocess.py -h` for the full list.

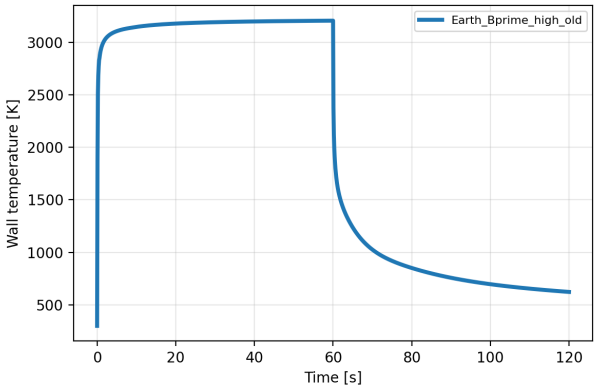


Figure 3. Wall temperature.

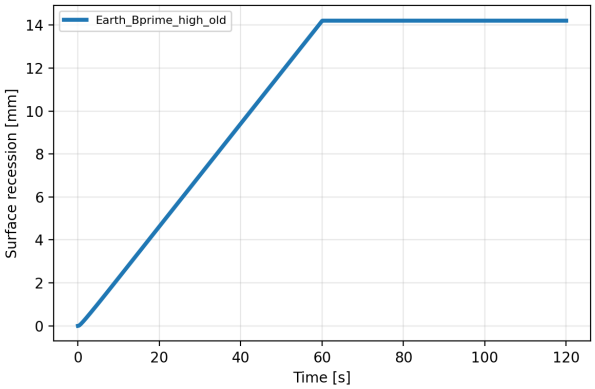


Figure 4. Surface recession.

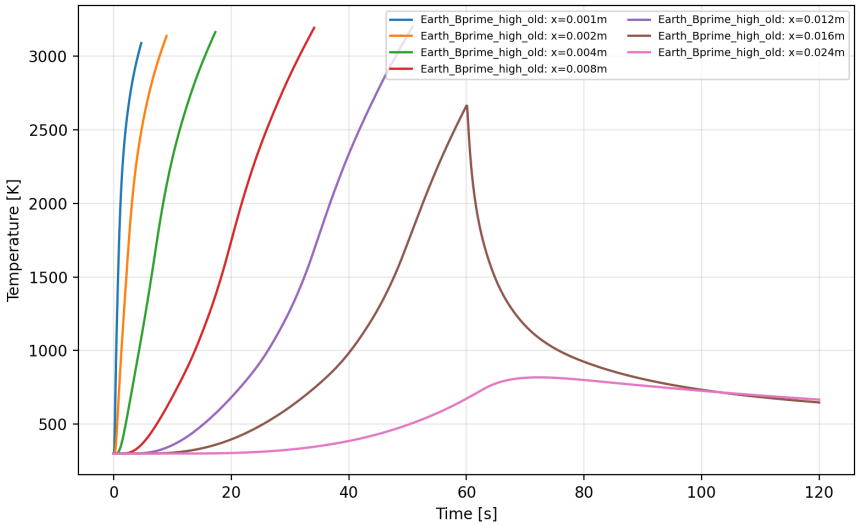


Figure 5. Temperature histories at fixed depths.

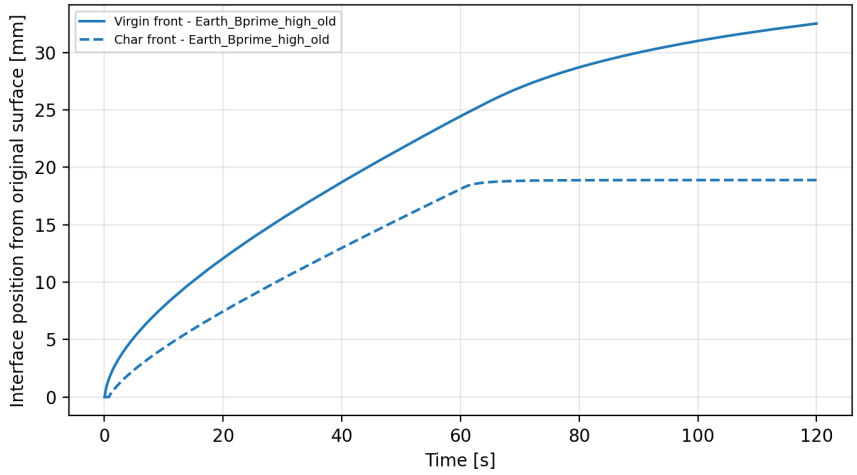


Figure 6. Positions of the virgin and char fronts.

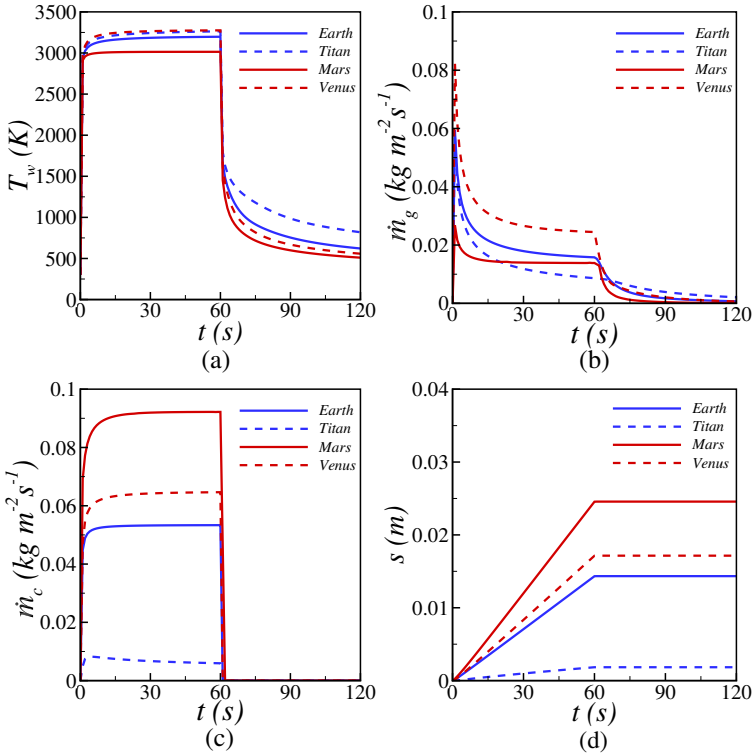


Figure 7. Response in the four atmospheres, high heating: (a) wall temperature, (b) pyrolysis-gas flux, (c) char flux, (d) recession.

What the simulation writes

The simulation output files are plain text. The main ones are listed in Table 4.

Table 4. Files in the case folder.

File	Columns
`out_<t>.dat`	x [m] (from the original surface), T [K], pressure [Pa], density [kg/m^3]
`boundarydat.dat`	t, pressure at the front node, wall temperature T_w [K], B'_g, B'_c, h_w [J/kg]
`BC.dat`	t, recession rate [m/s], front-face position [m]
`mgeq.dat`	t, pyrolysis-gas mass flux at the wall, char mass flux at the wall [kg/m^2/s]
`WEnergyBalance.dat`	t, then the terms of the surface energy balance [W/m^2] (listed in the README)
`inflocv.dat`, `inflocc.dat`	t, density threshold, position of the virgin front (98 % virgin) and of the char front (2 % char remaining)
`info.dat`	run header (environment, heating level, ramp times, settings) and final thickness

Part 3: Change the ramp

Ramp times

Use the flags `-tup`, `-toff`, `-tdown`, or edit `tRampUp`, `tRampOff`, `tRampDown` in `params.idp`. Set `-tend` larger than `-toff` + `-tdown` if you want the cooling phase.

```
# slower ramp: 1 s ramp-up, plateau until 4 s, 1 s ramp-down, cooling until 8 s
FreeFem++ -nw main.edp -env Earth -tup 1 -toff 4 -tdown 1 -tend 8 -ele 100
```

Ramp times of 0.1 s and longer were tested. For a faster ramp, reduce `-dt` and check that the run stays stable.

A new heating level

Add one line to the list in `params.idp` (section 2), then run with `-heat medium`:

```
if (heatbc == "low") HrPeak = 1.5e6;
else if (heatbc == "intermediate") HrPeak = 1.33e7;
else if (heatbc == "high") HrPeak = 2.5e7;
else if (heatbc == "medium") HrPeak = 8.0e6;   // new level [J/kg]
```

Going further

- **Different pulse shape:** replace the macro `pulse(t)` in `Datafolder/bc/Boundary1.idp`. It must return a number between 0 and 1 for every time (the file is shared by all three tutorials).
- **New environment:** add a table named `Bprime_table_modified_<Name>.dat` to `Datafolder/tacotdata/Bprimetables/`, add its wall pressure and number of rows to `params.idp` (for example `else if (Environment == "MyPlanet") { prwall = 5.0e4; nBprime = 61004; }`) and run with `-env MyPlanet`. The table layout is described in the README.
- **Tables off:** with `useAblation = false` the tables are not used and the sample does not recede. All environments then give almost the same result, so do not use this setting to compare environments.

If something goes wrong

Message or symptom	What to do
`Load error : msh3`	FreeFEM has no `msh3` plugin. See the Ubuntu / Debian note in “Before you start”.
`Error opening file ../../Datafolder/...`	FreeFEM was started outside the tutorial folder. `cd` into `Tutorials/ConvectiveRamp_Bprime` first.
`Sorry error in Optimization ...` or `nan`	The run diverged. Use `-dt 0.005` or smaller.
`ERROR: unknown Environment ...`	Misspelled name (capital letters matter), or it is not defined in `params.idp`.
The post-processing script cannot find the files	Check the case-folder name and that the run has finished. Run the script from the tutorial folder.
The run is very slow	Full runs (120 s, 500 elements) can take a few hours. Use the quick test first and keep `ploton = false`.

Quick reference

Table 5. Command-line flags. Each one overrides the matching line of `params.idp`.

Flag	Overrides	Default	Values
`-env`	`Environment`	Earth	Earth, Mars, Venus, Titan
`-heat`	`heatbc`	high	low, intermediate, high
`-tend`	`tend`	120	end time [s]
`-ele`	`ele`	500	number of elements
`-dt`	`dt`	0.005	time step [s]; keep ≤ 0.005
`-tup`	`tRampUp`	0.1	ramp-up time [s]
`-toff`	`tRampOff`	60	time when the ramp-down starts [s]
`-tdown`	`tRampDown`	0.1	ramp-down time [s]

References

[1] Frédéric Hecht. New development in freefem++. *Journal of Numerical Mathematics*, 20(3–4):1–4, 2012. https://doi.org/10.1515/jnum-2012-0013.

[2] Jean Lachaud and Nagi N Mansour. Porous-material analysis toolbox based on openfoam and applications. *Journal of Thermophysics and Heat Transfer*, 28(2):191–202, 2014. https://doi.org/10.2514/1.T4262.

[3] Jean R Lachaud, Alexandre Martin, Tom van Eekelen, and Ioana Cozmuta. Ablation test-case series# 2. numerical simulation of ablative-material response: Code and model comparisons. 2012. https://jeanlachaud.com/?page_id=132.